# Tanmay Bhuskute

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## SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience writing Go and Python for cloud infrastructure, REST APIs, and Docker/Kubernetes automation at Siemens. Current research optimizes LLM inference pipelines for latency and cost via model benchmarking and caching.

## EDUCATION

**Master of Science in Computer Science** Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis.

**Bachelor of Engineering in Computer Engineering** Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Machine Learning, Object-Oriented Programming.

## TECHNICAL SKILLS

**Programming:** Go, Python, Java, JavaScript, TypeScript, C/C++, SQL.

**Web & Backend:** REST APIs, Node.js, FastAPI, Flask, Django, React, HTML/CSS.

**Databases:** PostgreSQL, MySQL, Redis.

**Developer Tools:** Docker, Kubernetes, AWS, GitLab CI/CD, Jenkins, ArgoCD, Helm, Terraform, Ansible, Linux, Bash, SonarQube, Google Cloud Platform, Azure.

**AI & Data:** LLM inference optimization, model benchmarking, quantization, caching strategies, Gemini, Vertex AI, PyTorch, scikit-learn, LangChain, pandas, NumPy.

## WORK EXPERIENCE

**Graduate Researcher** | University of Cincinnati Feb 2026 – Present

- Built two-stage memory-augmented LLM inference pipeline decoupling video understanding from device automation to cut redundant model calls, showing **25.7x latency reduction** (39.4s vs 1012.1s avg per run) and lower per-run compute cost while preserving task fidelity.
- Ran provider-agnostic model benchmarking (Gemini, GPT-4o, Llama, Qwen, MiniCPM) against the **ViBR** baseline across 53 runs on 20+ apps with a config-driven harness for reproducible evaluation.

**Software Development Engineer** | Siemens Industry Software Jul 2022 – Jun 2024

- Wrote Python (Boto3) automation for cloud infrastructure workflows and REST APIs, replacing legacy Ruby scripts and cutting deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months against a live fleet, diagnosing infrastructure issues across Linux, EC2, and databases with no service interruptions on mission-critical workloads.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault; designed Ansible Blue-Green framework cutting upgrade downtime by **50%**.
- Served as Quality Coach and SonarQube Specialist, raising automated code coverage from **35% to 78%**; mentored **5 engineers** and folded review feedback into shared coding standards.

**Software Developer Intern** | Siemens Industry Software Mar 2022 – Jun 2022

- Automated AWS resource provisioning with **Terraform**, reducing deployment time by **30%**.
- Built **Go + Terratest** validation framework, catching **95%** of misconfigurations pre-production.

**Software Developer Intern** | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management and automated manual processes, improving operational efficiency by **40%**.

## PROJECTS AND RESEARCH

**PolicyPilot – Policy Intelligence Portal** [Python, TS, Node, Gemini, MCP, Docker] May 2026

- Architected evidence-grounded **RAG inference pipeline** over Gemini 2.5-flash; benchmarked hallucination rates to pick the model and built a **17-tool LLM API server** for external AI agents.

**NYC Mobility Forecasting** [Python, PySpark, Delta Lake, scikit-learn, FastAPI, Redis] Jul 2026

- Built ML system forecasting demand across **263 zones** (**2.3M rows**); Poisson hurdle model improved MAPE **18.77% to 16.52%**, served via a FastAPI REST API with Redis caching in Docker.

## EXTRACURRICULAR ACTIVITIES

**Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).